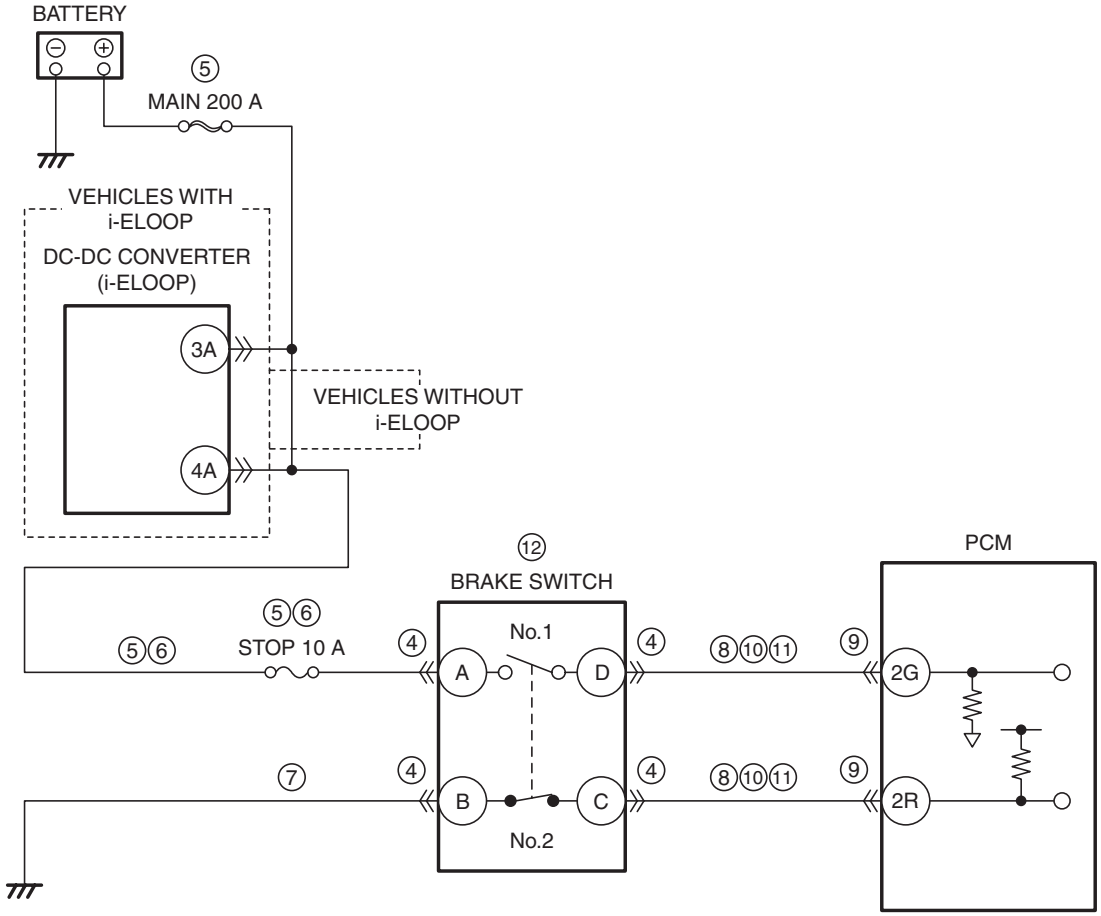
DTC P0504:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

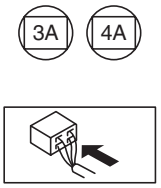
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DTC P0504:00	Brake switch circuit problem
DETECTION CONDITION	• The condition in which the brake switch No.1 and No.2 signals are both on or off for **3 s or more** is repeated five times when performing the brake operation. **Diagnostic support note** • This is a continuous monitor (other). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	**Caution** **• Inspect the brake switch with it installed to the brake pedal, otherwise the brake switch may not operate normally. If the brake switch is removed from the brake pedal, replace the brake switch with a new one.** • Brake switch connector or terminals malfunction • Short to ground or open circuit in brake switch No.1 power supply circuit (without i-ELOOP) — Short to ground in wiring harness between MAIN 200 A fuse and brake switch terminal A — MAIN 200 A fuse and/or STOP 10 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and brake switch terminal A • Short to ground or open circuit in brake switch No.1 power supply circuit (with i-ELOOP) — Short to ground in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and brake switch terminal A — STOP 10 A fuse malfunction — Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and brake switch terminal A • Open circuit in wiring harness between brake switch terminal B and body ground • Short to ground in wiring harness between the following terminals: — Brake switch terminal D—PCM terminal 2G — Brake switch terminal C—PCM terminal 2R • PCM connector or terminals malfunction • Short to power supply in wiring harness between the following terminals: — Brake switch terminal D—PCM terminal 2G — Brake switch terminal C—PCM terminal 2R • Open circuit in wiring harness between the following terminals: — Brake switch terminal D—PCM terminal 2G — Brake switch terminal C—PCM terminal 2R • Brake switch malfunction • PCM malfunction

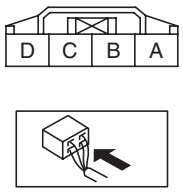
DTC P0504:00	Brake switch circuit problem



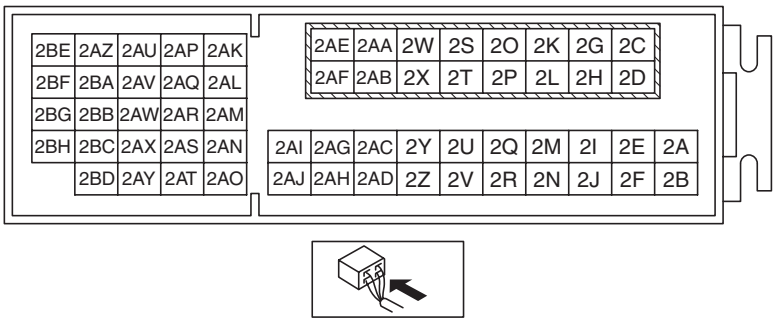
DC-DC CONVERTER (i-ELOOP) WIRING HARNESS-SIDE CONNECTOR



BRAKE SWITCH WIRING HARNESS-SIDE CONNECTOR



PCM WIRING HARNESS-SIDE CONNECTOR



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the PENDING CODE/DTC P0703:00 also present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC P0703:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
4	**INSPECT BRAKE SWITCH CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the brake switch connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 13.
		No	Vehicles with i-ELOOP: • Go to Step 6. Vehicles without i-ELOOP: • Go to the next step.
5	**INSPECT BRAKE SWITCH NO.1 POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the brake switch connector is disconnected. • Measure the voltage at the brake switch terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to Step 7.
		No	Inspect the MAIN 200 A fuse and STOP 10 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and brake switch terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the malfunctioning fuse. • If the fuse is damaged: — Replace the malfunctioning fuse. • If all fuses are normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and brake switch terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 13.

STEP	INSPECTION		ACTION
6	**INSPECT BRAKE SWITCH NO.1 POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the brake switch connector is disconnected. • Measure the voltage at the brake switch terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Remove the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the STOP 10 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between STOP 10 A fuse and brake switch terminal A. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and brake switch terminal A. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Go to Step 13.
7	**INSPECT BRAKE SWITCH NO.2 GROUND CIRCUIT FOR OPEN CIRCUIT** • Verify that the brake switch connector is disconnected. • Inspect for continuity between brake switch terminal B (wiring harness-side) and body ground. • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between brake switch terminal B and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Inspect for the following: — Open circuit between brake switch and body ground — Loose or lifting ground point • Repair or replace the malfunctioning part. Go to Step 13.

STEP	INSPECTION		ACTION
8	**INSPECT BRAKE SWITCH CIRCUIT FOR SHORT TO GROUND** • Verify that the brake switch connector is disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Brake switch terminal D — Brake switch terminal C • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — Brake switch terminal D—PCM terminal 2G — Brake switch terminal C—PCM terminal 2R **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to Step 13.
		No	Go to the next step.
9	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 13.
		No	Go to the next step.
10	**INSPECT BRAKE SWITCH CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the brake switch and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the following terminals (wiring harness-side): — Brake switch terminal D — Brake switch terminal C • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Brake switch terminal D—PCM terminal 2G • Brake switch terminal C—PCM terminal 2R **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 13.
11	**INSPECT BRAKE SWITCH CIRCUIT FOR OPEN CIRCUIT** • Verify that the brake switch and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between the following terminals (wiring harness-side): — Brake switch terminal D—PCM terminal 2G — Brake switch terminal C—PCM terminal 2R • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Brake switch terminal D—PCM terminal 2G • Brake switch terminal C—PCM terminal 2R **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 13.
12	**INSPECT BRAKE SWITCH** • Inspect the brake switch. (See BRAKE SWITCH INSPECTION.) • Is there any malfunction?	Yes	Replace the brake switch, then go to the next step. (See BRAKE PEDAL REMOVAL/INSTALLATION [L.H.D.].) (See BRAKE PEDAL REMOVAL/INSTALLATION [R.H.D.].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
13	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Switch the ignition ON (engine off). • After depressing the brake pedal for **3 s or more**, perform the operation of depressing and releasing the brake pedal **5 times**. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
14	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.